

- 4 The first law of thermodynamics may be written as

$$\Delta U = Q + W$$

where W is the work done on a system.

- (a) State the meaning, in this equation, of the symbols ΔU and Q .

ΔU

Q

[2]

- (b) An ideal gas at a pressure of $2.0 \times 10^5 \text{ Pa}$ has an initial volume of 0.048 m^3 , an initial thermodynamic temperature of 560 K , and an initial internal energy of 14.4 kJ .

The volume of the gas is reduced at constant pressure to a final volume of 0.012 m^3 .

- (i) Show that the magnitude of the work done on the gas is 7200 J .

[2]

- (ii) Explain whether the work done on the gas is positive or negative.

..... [1]

- (iii) Explain why the final internal energy of the gas is 3.6 kJ .

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.....

..... [2]

- (c) For the reduction of the volume of the gas in 4(b), calculate Q . Give a unit with your answer.

$Q =$ unit [2]

[Total: 9]